Peixuan Yang

Incoming Year 2 BEng (Hons) · Computer Science and Electronic Engineering

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Research interests AI hardware–world-model co-design; photonic computing and computer architecture; efficient generative inference; trustworthy autonomous and multi-robot systems.

EDUCATION

University of Liverpool

BEng (Hons), Computer Science and Electronic Engineering | Sep 2026 – Jun 2028 (expected)

XJTLU 2+2 pathway; Years 2–3 at Liverpool.

Xi'an Jiaotong-Liverpool University

Computer Science and Technology · Suzhou, China | Sep 2024 – Jul 2026

- 2025/26 weighted average: 71% (First-Class level). Selected marks: Computer Systems 88%; Engineering Mathematics II 86%; Digital Electronics I 76%; Data Structures and Algorithms 76%.

SELECTED PUBLICATION

Y. Wang, Y. Gao, **P. Yang**, and B. Zhao. "Optimized Fine-tuning and Pseudo-Data Strategies for Cross-Domain Low-Resource Language Cantonese-English Neural Machine Translation." 11th International Conference on Robotics and Artificial Intelligence (ICRAI), Nagoya, Japan, 2025.

Co-second author and presenter. Best Presentation Award.

RESEARCH EXPERIENCE

PhotonWM · MRR Acceleration for Autoregressive World Models

Student Researcher · Supervisor: Prof. Jiaqi Liu | Jul 2026 – Present

- Co-designed a hybrid photonic–electronic architecture mapping fixed-weight Linear operators to 64 × 64 microring-resonator arrays, with dynamic attention, nonlinearities, sampling and control retained on the electronic backend.
- Reproduced and profiled the official LPWM BAIR-128 checkpoint on an RTX 5090; built operator and tensor-shape ledgers and identified recurrent Dynamics execution as 91.56% of measured stage latency.
- Integrated measured traces with LLMCompass, SimPhony and event-driven scheduling to evaluate weight residency, MRR programming, conversion, latency, energy and rollout quality. Contributed evaluation methodology and manuscript revision.

Full manuscript submitted to DATE 2027.

SURF 2026 · Collaborative Proof-of-Sensing and Multi-Robot Auditing

Summer Undergraduate Research Fellow · Supervisor: Prof. Limin Yu | Jun – Aug 2026

- Conducted largely independent literature-to-prototype research on CPoS and lightweight trusted auditing for UAV, surface and underwater robot missions.
- Implemented and tested a local Hardhat EVM network, Solidity sensing registry and ethers.js client for digest submission, retrieval and audit events; documented transaction and gas evidence while keeping raw sensing data off-chain.
- Built a three-X500 ROS 2 Jazzy / Gazebo Harmonic / PX4 v1.17 simulation environment with separate identities and namespaces; validated take-off, hover, waypoint, return and landing. Specified a next-stage protocol for signed multi-source mission evidence and future physical-sensing verification.

Programme completed; SURF certificate issued September 2026.

Low-Resource Cantonese–English Neural Machine Translation

Researcher · Co-second author and presenter, ICRAI 2025 | Jun 2025 – Jan 2026

- Constructed and cleaned a 1.1-million-pair parallel corpus using syntactic filtering, normalisation, deduplication and LaBSE-based semantic alignment.
- Fine-tuned NLLB-200-distilled-600M and designed a nested pseudo-data sampling strategy; the best 200k:200k pseudo-to-real setting achieved BLEU 29.63 and chrF 56.65.

PROFESSIONAL EXPERIENCE

Research Institute of Tsinghua University in Shenzhen

R&D Intern · Marine Hydrogen Energy R&D Center | May – Jul 2026

- Completed a two-month R&D placement; the official internship certificate commends initiative, collaboration and strong overall performance.

Wuhan University City Hospital Co., Ltd.

Medical Data Analysis / IT R&D Intern · Wuhan, China | Jan – Mar 2026

- Used Python to process and analyse operational data, supported technical troubleshooting and contributed to internal healthcare IT workflows.
- Evaluated AI-assisted troubleshooting and organised internal knowledge-base content for more consistent technical support.

Wuhan Dongfangyunjin Information Technology Co., Ltd.

Backend and UI Programmer (Medical Devices) · Wuhan, China | Feb – Dec 2023

- Developed web interfaces and external API integrations for secure data exchange and partner bidding information; implemented RBAC user management with permission and account CRUD operations.

SELECTED TECHNICAL PROJECT

HiAgent Intelligent HR System

Lead Knowledge Base Engineer · XJTLU Hackathon

- Designed a retrieval-augmented candidate-evaluation workflow; curated job descriptions, CVs and interview records for vector retrieval and LLM interpretation.
- Integrated PDF/Word parsing with job-description analysis, candidate scoring and dynamic interview-question generation.

TECHNICAL SKILLS

Research and ML Python, PyTorch, model profiling, workload tracing, reproducible experiments, NLLB/LLM fine-tuning.

AI systems and hardware LPWM, world models, LLMCompass, SimPhony, microring-resonator photonic computing, hybrid photonic–electronic architectures.

Robotics and security ROS 2, PX4 SITL, Gazebo, Solidity, Hardhat, ethers.js, blockchain and permissioned-ledger prototyping.

Software Git, Linux, Java, SQL, REST APIs, LaTeX.

ADDITIONAL TRAINING AND HONOURS

CEAIE Innovative Talents Science Training Programme

Machine Learning, Network Security and Large Language Models | Mar – May 2025

Instructed by Prof. Pietro Liò (University of Cambridge); final grade 97.40 / A+.

- Best Presentation Award — ICRAI 2025, Nagoya, Japan.
- Jiangsu Province Anti-Drug Promotion Ambassador and Outstanding Anti-Drug Youth.